25900364 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACECT200, MG3VACECC200 - GREEN ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question

Answer all questions

Each question carries **1** marks

1. Cadmium exposure can cause damage to the: [K] / [CO1]

   a). Kidneys b). Heart

   c). Skin d). Eyes

2. Which of the following combined techniques shows high efficiency for aluminum recovery from PCB [A] / [CO2]

   a). Direct smelting + electrowinning b). Mechanical separation + hydrometallurgical treatment

   c). Pyrolysis + magnetic separation d). Bioleaching + flotation

3. Which is an environmental concern of chemical recycling? [A] / [CO3]

   a). High energy recovery b). Toxic effluents

   c). Improved purity of products d). Low carbon footprint

4. Which is a limitation of mechanical recycling compared to chemical recycling? [U] / [CO3]

   a). Limited recyclability after repeated cycles b). High purity output

   c). High cost d). Low scalability

5. Why is e-waste recycling considered a “double benefit” process? [U] / [CO1]

a). It increases jobs and increases pollution
b). It reduces pollution and recovers valuable resources
c). It reduces metals and reduces resources
d). It only saves land space

6. Why are furans and dioxins particularly dangerous? [U] / [CO1]

a). They are water-soluble
b). They persist long and cause cancers
c). They improve plant growth
d). None

7. The use of resin, gelatin, and shellac in green electronics mainly supports [U] / [CO2]

a). Disposable, eco-friendly devices with minimal e-waste
b). Heavy-duty aerospace applications
c). High-voltage power electronics
d). Nuclear reactor control systems

8. To improve recovery of plastics from mixed e-waste, which method should be applied after shredding? [U] / [CO3]

a). Pyrolysis
b). Density-based separation
c). Magnetic separation
d). Electro-refining

9. Long-term soil exposure to e-waste leads to: [U] / [CO1]

a). Better irrigation
b). Increase in organic matter
c). Reduced fertility
d). More crop yield

10. E-waste can best be described as: [K] / [CO1]

a). Biodegradable plastics
b). Old electronic devices
c). Food and kitchen waste
d). Garden waste

11. Which of the following is NOT a core principle of the 3Rs in electronics? [A] / [CO2]

a). Reduce
b). Reuse
c). Reinvent
d). Recycle

12. Which of the following is a limitation of sustainable materials compared to traditional ones? [A] / [CO2]

a). Non-biodegradability
b). Lower carbon footprint in production
c). Poorer moisture resistance and durability
d). Infinite lifespan

13. Which method should be selected to complement magnetic separation for complete metal recovery from e-waste? [E] / [CO3]

a). Landfilling
b). Incineration
c). Manual dismantling
d). Eddy current separation

14. The first step in e-waste recycling is usually: [U] / [CO1]

a). Burning the waste b). Collection and segregation

c). Exporting abroad d). Throwing in landfills

15. Arsenic exposure causes: [U] / [CO1]

a). Skin lesions and cancers b). Hair loss

c). Diabetes d). None

16. One way to make electronic devices more sustainable is to: [A] / [CO2]

a). Use renewable energy in manufacturing b). Burn old devices

c). Use more plastic d). Avoid repairing devices

17. Which of these is a cost advantage of mechanical recycling? [An] / [CO3]

a). High setup costs b). Low operational costs

c). High chemical costs d). Frequent catalyst replacement

18. Which cellulose derivative is commonly used in pharmaceuticals? [K] / [CO2]

a). Cellulose acetate b). Nitrocellulose

c). Carboxymethyl cellulose (CMC) d). Cellulose nitrate

19. What happens to aquatic biodiversity in heavily e-waste-polluted zones? [U] / [CO1]

a). Species diversity reduces b). Species diversity increases

c). No change d). Only algae flourish

20. Protein-based dielectrics are typically derived from: [K] / [CO2]

a). Petrochemical sources b). Natural sources such as plants and animals

c). Metallic compounds d). Fossil fuels

21. Which plastics are the hardest to separate using gravity methods? [U] / [CO3]

a). PET and PVC b). ABS and PS

c). PP and PE d). PC and nylon

22. Children are more vulnerable to organic toxins because: [U] / [CO1]

a). Higher metabolic rate and developing organs b). They play outdoors only

c). They do not eat electronics d). None

23. In copper recovery, aqua regia is avoided for large-scale PCB recycling because: [A] / [CO2]

a). It is too cheap b). It cannot dissolve copper

c). It produces toxic fumes and is corrosive d). It requires high temperature

24. Identify the primary advantage of in-situ leaching. [U] / [CO3]

a). Low capital costs b). High recovery rates

c). Minimal environmental impact d). All of the above

25. Which of the following makes recycling of e-waste costly? [K] / [CO1]
a). High labor wages b). Lack of plastic in devices
c). Complex mixture of materials d). Abundant recycling units

26. Which of the following best describes the challenge of e-waste management? [U] / [CO1]
a). Safe, cheap, and quick process b). Easy, simple, and eco-friendly
c). Complex, hazardous, and costly process d). No effect on health or environment

27. Which material is mentioned as an eco-friendly dielectric layer derived from silk? [U] / [CO2]
a). Polyethylene b). Silk-based substrates
c). Aluminum foil d). Copper wire

28. Select the main benefit of electrochemical conversion in metal recovery. [U] / [CO3]
a). High efficiency b). Low energy consumption
c). Selective metal recovery d). All of the above

29. Lead accumulation in plants results in: [U] / [CO1]
a). Enhanced flowering b). Stunted growth and leaf damage
c). Resistance to drought d). None

30. Which of the following does NOT belong to medical E-waste? [K] / [CO1]
a).X-ray machine b).Mobile phone
c).ECG machine d).MRI scanner

31. Gelatin is particularly useful in green electronics because it is: [A] / [CO2]
a). Conductive by nature b). Biodegradable and forms flexible films
c). Completely water-proof d). Derived from metals

32. Identify the type of pyrolysis that occurs at moderate temperatures and residence times. [U] / [CO3]
a). Fast pyrolysis b). Intermediate pyrolysis
c). Slow pyrolysis d). Torrefaction

33. Hospitals create e-waste mainly through: [U] / [CO1]
a). Diagnostic and monitoring devices b). Food supplies
c). Bedsheets d). Medicines only

34. Recycling helps in reducing: [A] / [CO2]
a). Air pollution b). Landfill waste
c). Energy consumption d). All of the above

35. During mechanical recycling, why is a uniform feed size important before the shredding process? [U] / [CO3]

a). It reduces the need for downstream sorting

b). It prevents uneven wear and tear on shredder blades

c). It increases the magnetic properties of metals

d). It improves the color of recovered plastics

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. Differentiate between hazardous and non-hazardous compounds in e-waste with examples. [K] / [CO1]

37. What is bioleaching, and how does it help in gold recovery from e-waste? [K] / [CO2]

38. Discuss one major limitation of paper-based dielectric materials in electronics. [A] / [CO2]

39. Evaluate the risks of applying air classification before shredding. [E] / [CO3]

40. What components of a PC can be reused? Explain briefly. [U] / [CO4]

[5x3 = 15]